

MSc Project 1: Time-Series-Enhanced Predictive Process Monitoring Final Report

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Abstract. Keywords: Predictive Process Monitoring · Process Mining
· Time-Series Features · Machine Learning · Model Evaluation · Explain-
ability

1 Introduction

2 Data and Preprocessing

2.1 Event Log: BPI Challenge 2017

2.2 Preprocessing Pipeline

2.3 Temporal Train-Test Split

3 Feature Engineering

3.1 Log-Based Features (Baseline)

3.2 Time-Series Features (Enhanced)

3.3 Feature Extraction Pipeline

4 Model Specification and Evaluation Protocol

4.1 Prediction Tasks

- **Outcome prediction (binary classification):** predict whether a case will end in a deviant state vs regular completion. Decayed AUC-PR.
- **Remaining-time prediction (regression):** predict time (in hours) until case completion. Decayed MAE.

4.2 Models

4.3 Hyperparameter Optimisation

4.4 Prefix-Length Selection

4.5 Evaluation Metrics

- **Classification (outcome):** AUC-PRC, AUC-ROC, F1-score.
- **Regression (remaining time):** MAE, RMSE, (maybe) MAPE.
- **Decayed metrics:** weight more recent prefixes higher.

4.6 Baseline vs Enhanced Comparison

5 Results

5.1 Outcome Prediction

5.2 Remaining-Time Prediction

5.3 Feature Importance Analysis

5.4 Ablation: Which Time-Series Features Matter?

6 Discussion

6.1 Do Time-Series Features Improve Predictive Performance?

6.2 Why Time-Series Features Help (or Do Not)

6.3 Computational Overhead

6.4 Generalisability and Limitations

7 Threats to Validity

8 Conclusion

8.1 Summary

Data and Code Availability

References

1. van Dongen, B.F.: BPI Challenge 2017. Eindhoven University of Technology (2017). <https://research.tue.nl/en/datasets/bpi-challenge-2017>